Formative Assessment - Corrections

MYP 3 Mathematics - Laws of Exponents - correct worked answers

Each question shows the correct working step by step. **Red** notes show the mistake made in the original test.

Level 1-2

1. Match each law of exponents with its name.

Law	Name	Letter
$a^m \times a^n = a^{m+n}$	Product law	**C**
$\frac{a^m}{a^n} = a^{m-n}$	Quotient law	**A**
$(a^m)^n = a^{m \times n}$	Power of a power law	**E**
$a^{-n} = \frac{1}{a^n}$	Negative exponent law	**D**
$a^0 = 1$	Zero exponent law	**B**

Answer: 1-C, 2-A, 3-E, 4-D, 5-B

2. (a) Find the value of 5^{-3}

Negative -> move to bottom	$\frac{1}{5^3} = \frac{1}{5 \times 5 \times 5}$
Answer:	**1/125**

Mistake made: stopped at $1/5^3$ - always finish the calculation.

. (b) Find the value of 9^0

Any number to the power 0 is 1	
Answer:	**1**

3. Select the correct simplification.

(a) add: 4 + 3 = 7	$x^4 \times x^3 = x^7$
(b) subtract: 6 - 2 = 4	$\frac{y^6}{y^2} = y^4$
(c) multiply: 2 x 5 = 10	$(z^2)^5 = z^{10}$
Answer:	**x^7 ; y^4 ; z^{10}**

Level 3-4

4. (a) Find the value of -3^{-3}

The minus is outside; the power is only on 3	$-\left(\frac{1}{3^3}\right)$
Answer:	**-1/27**

. (b) Find the value of $\left(\frac{1}{3}\right)^{-3}$

Negative exponent flips the fraction	$\left(\frac{3}{1}\right)^3 = 3^3$
Answer:	**27**

. (c) Find the value of $\left(\frac{6}{7}\right)^{-2}$

Flip the fraction	$\left(\frac{7}{6}\right)^2 = \frac{7^2}{6^2}$
Answer:	**49/36**

Mistake made: wrote $7/6^2$ - the power must go on BOTH the top and the bottom.

. (d) Find the value of $-9^0 + (-9)^0$

-9^0: power only on 9	$-(1) = -1$
$(-9)^0$: whole bracket	1
Add	$-1 + 1$
Answer:	**0**

Mistake made: worked out -9 instead of using the zero exponent law on both parts.

5. (a) Simplify $3^{-2}x^{-2}y^3z^{-11}t^0$

$t^0 = 1$	
Negatives move to the bottom	$\frac{y^3}{3^2x^2z^{11}}$
Answer:	**y^3 / ($9x^2z^{11}$)**

Mistake made: left blank - every negative exponent moves to the bottom; t^0 becomes 1.

. (b) Simplify $(2x^3)^4$

Power goes to the 2 AND to x^3	$2^4 \times x^{3 \times 4}$
Answer:	**$16x^{12}$**

Mistake made: wrote $8x^{12}$ - 2^4 = 2x2x2x2 = 16.

. (c) Simplify $4r^{-3} \times 2r^{2}$

Numbers: 4 x 2 = 8	
r: -3 + 2 = -1	$8r^{-1} = \frac{8}{r}$
Answer:	**8/r**

Mistake made: wrote 8r - the exponent is -1, so r moves to the bottom.

. (d) Simplify $(2x^{4}y^{-3})^{-1}$

Multiply every exponent by -1	$2^{-1}x^{-4}y^{3}$
Negatives move to the bottom	$\frac{y^{3}}{2x^{4}}$
Answer:	**y^3 / ($2x^4$)**

Level 5-6

6. Simplify $\frac{3x^{-2}y^3}{6xy^{-4}}$

Numbers: 3 / 6	$\frac{1}{2}$
x: -2 - 1 = -3	x^{-3}
y: 3 - (-4) = 7	y^7
Move x^{-3} to the bottom	$\frac{y^7}{2x^3}$
Answer:	**y^7 / ($2x^3$)**

Mistake made: subtracted the numbers (6 - 3) instead of dividing them, and lost the y exponent.

7. Simplify $\frac{x^2y^{-1}}{z^3} \times \frac{(xz)^2}{y^4}$

Open the bracket	$(xz)^2 = x^2z^2$
Multiply tops and bottoms	$\frac{x^2 \cdot x^2 \cdot z^2 \cdot y^{-1}}{z^3 \cdot y^4}$
x: 2 + 2 = 4	x^4
y: -1 - 4 = -5	$\frac{1}{y^5}$
z: 2 - 3 = -1	$\frac{1}{z}$
Answer:	**x^4 / (y^5z)**

Mistake made: cross-cancelled in a multiplication and did not square the bracket.

8. Simplify $\frac{m^3n^2p}{p^4q} \div \frac{m^5n^2}{(pq)^3}$

Keep - Change - Flip	$\frac{m^3n^2p}{p^4q} \times \frac{p^3q^3}{m^5n^2}$
m: 3 - 5 = -2	$\frac{1}{m^2}$
n: 2 - 2 = 0	$n^0 = 1$
p: (1 + 3) - 4 = 0	$p^0 = 1$
q: 3 - 1 = 2	q^2
Answer:	**q^2 / m^2**

Mistake made: did not flip the second fraction before multiplying.

9. A student simplified $\frac{(2a^3)^2}{4a^{-2}}$

(a) Step 1 mistake:	
The 2 was not squared	$(2a^3)^2 = 4a^6$ not $2a^6$
Step 2 mistake:	
6 - (-2) is 8, not 4	$\frac{a^6}{a^{-2}} = a^{6-(-2)} = a^8$
(b) Correct working	$\frac{4a^6}{4a^{-2}} = 1 \times a^{6+2}$
Answer:	**a^8**

Mistake made: said 2 x 3 = 6 for the number, and flipped the fraction instead of subtracting exponents.

Level 7-8

10. (a) A cube has edge length $4x^2y$ cm. Find its volume.

Volume of a cube = edge^3	$(4x^2y)^3$
Power goes to everything	$4^3 \times x^{2\times 3} \times y^3$
Answer:	**$64x^6y^3$ cm^3**

Mistake made: wrote 16xy - the edge must be cubed, and every part gets the power.

. (b) It is cut into small cubes of edge $2xy^{-1}$ cm. How many small cubes?

Small cube volume	$(2xy^{-1})^3 = 8x^3y^{-3}$
Number = big volume / small volume	$\frac{64x^6y^3}{8x^3y^{-3}}$
64 / 8 = 8 ; x: 6 - 3 = 3	$8x^3$
y: 3 - (-3) = 6	y^6
Answer:	**$8x^3y^6$ small cubes**

Mistake made: divided the edge lengths instead of the volumes.